

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India **09/2021 – 01/2025 (3.5 Years)**

- Designed RTL for high-speed **Ethernet MAC, PCS, and FEC IP cores**, supporting multi-rate architectures up to 1.6Tb, contributing to the development of next-generation AI infrastructure.
- Integrated digital design best practices, including **synthesis and timing closure**, into the IP delivery process, ensuring adherence to stringent performance specifications for high-speed networking.
- Collaborated daily with co-located design, verification, and back-end engineers to accelerate the development lifecycle of complex IP cores, reducing inter-team communication latency by an estimated **20%**.
- Supported production environments for a financial-services client, monitoring transaction flows and implementing operational alerting for failed messages, sustaining **99.9% uptime** for critical payment processing workflows.
- Automated data ingestion and reporting processes using **Python scripting** and SQL, reducing manual effort by **15 hours per week** and improving data accuracy for enterprise business systems.
- Developed and deployed feature enhancements for a Java Spring Boot-based payments operations dashboard, building **REST APIs** that facilitated seamless integration with downstream financial systems.

SKILLS

Hardware Description Languages – Verilog, SystemVerilog

Digital Design – RTL Design, State Machines, Clocking, Timing Closure, Synthesis

Networking Standards – Ethernet (IEEE 802.3dj), High-Speed Networking

Software Development – Java, Spring Boot, Python, REST APIs, SQL

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Observability & Monitoring – Splunk, Dynatrace, CloudWatch

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin **02/2026**

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India **01/2021**

PROJECTS

1.6Tbps Ethernet IP Controller for AI Infrastructure (HCL Technologies)

- Engineered **RTL designs** for high-speed Ethernet MAC, PCS, and FEC IP cores, enabling data transfer rates up to **1.6Tbps** to support AI and hyperscale computing demands.
- Implemented **digital design best practices**, including Verilog/SystemVerilog for RTL, synthesis, and timing closure, achieving sub-nanosecond signal integrity requirements.

- Collaborated with cross-functional teams (design, verification, back-end) in an agile environment to deliver configurable IP across multi-rate, multi-port architectures, reducing development cycles by **15%**.
- Contributed to the active development and specification of **IEEE 802.3 standards**, influencing the future of high-speed networking protocols.

Financial Transaction Monitoring & Reconciliation Dashboard (HCL Technologies)

- Developed and maintained a **payments operations dashboard**, integrating transaction monitoring and reconciliation workflows for a financial-services client.
- Implemented real-time operational alerting for failed and stuck payment messages, improving incident response times by **25%** and ensuring adherence to strict SLAs.
- Supported production environments with a focus on audit trails and **GDPR-aware** data handling, collaborating with business analysts for requirements and UAT.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)